

# Advancing 3D-IC Design Enablement Through R&D Innovation and Standards Development

CT Kao, Product Engineering Group Director

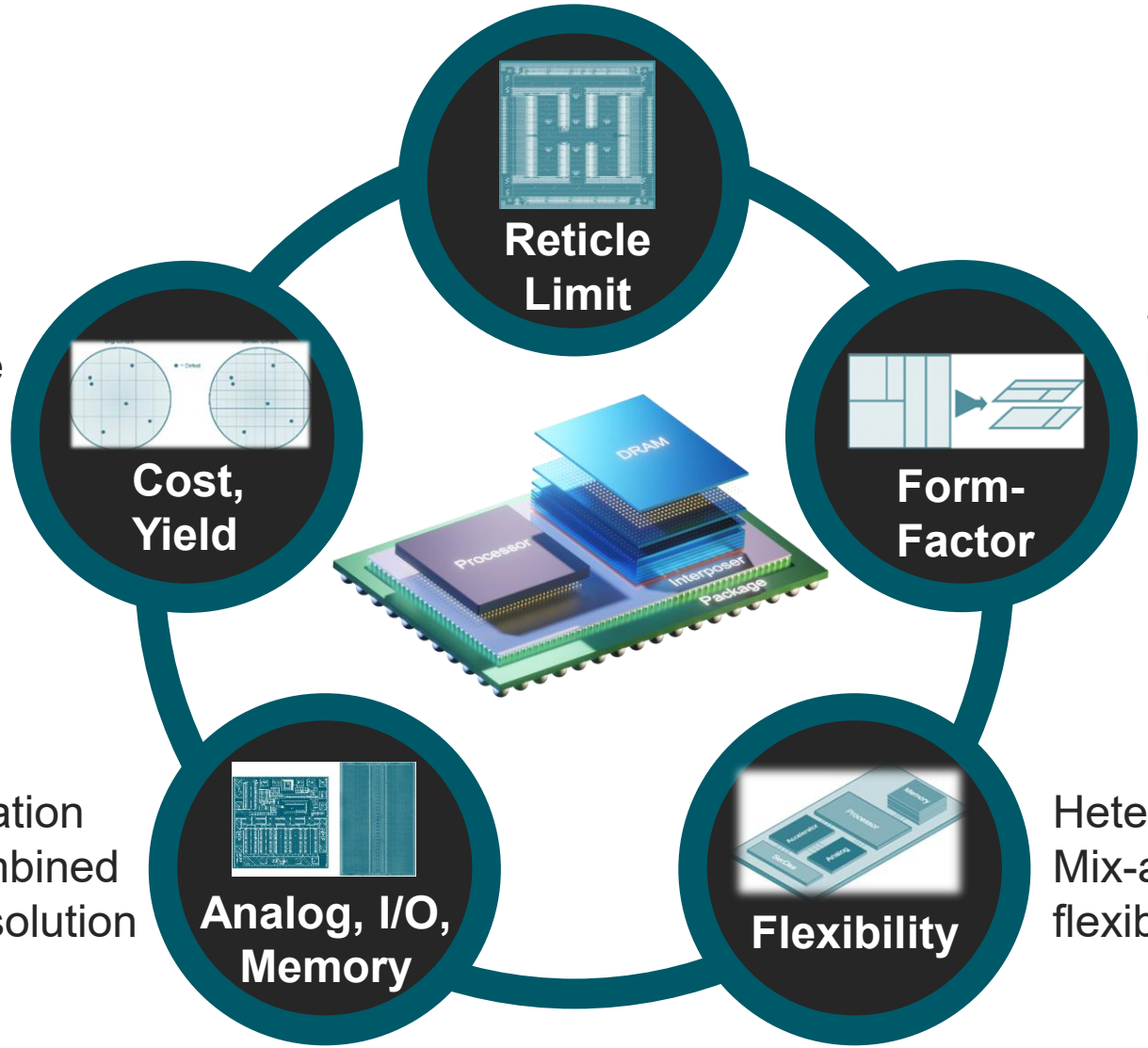
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# Why 3D-IC?

Physical limitation on die size

Poor yield / High cost for increasing die size

Size and form factor limited applications

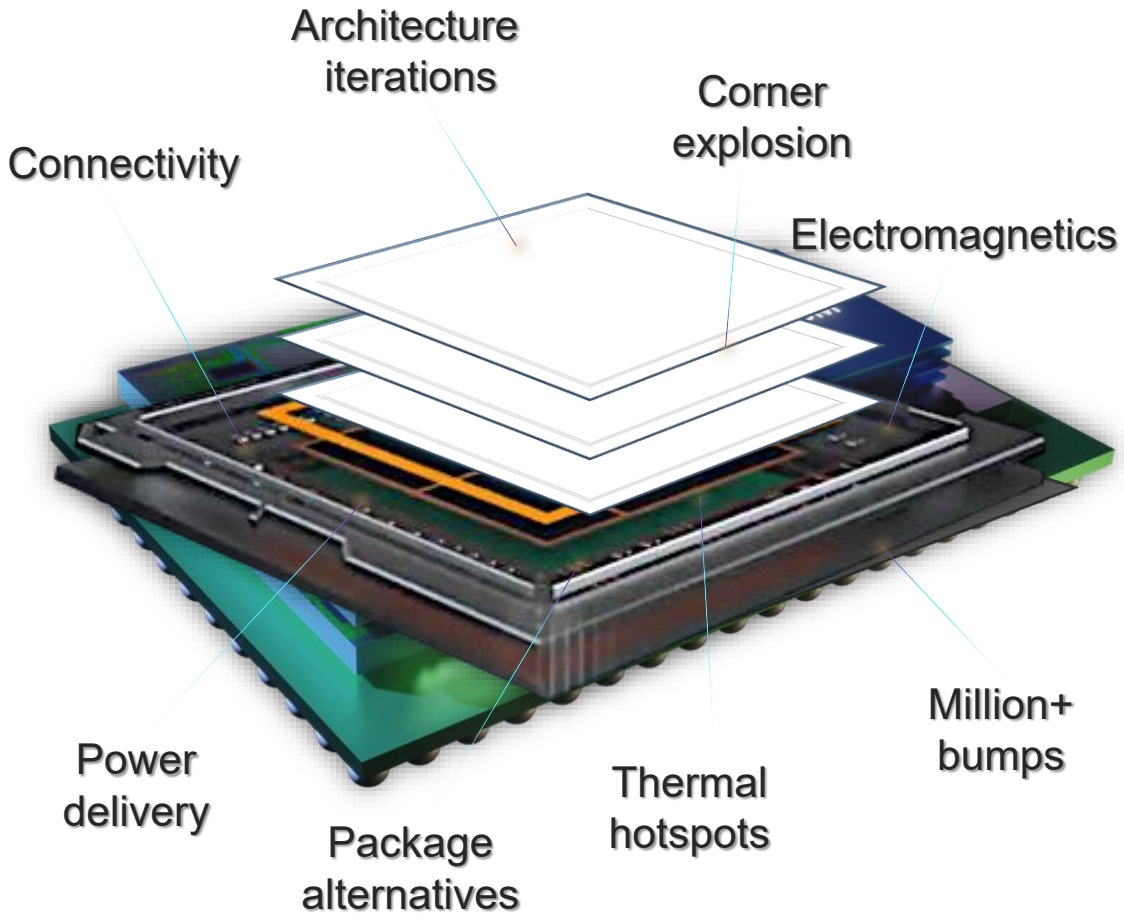



Enabling full utilization and benefit of combined advanced nodes solution

Heterogenous Integration: Mix-and-match provides flexibility with optimization


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# Multi-Chiplet Design Challenges and Solution




## Integrity™ 3D-IC Platform

3D System-Level Planning

Implementation

| Digital | Analog | RF |
|---|---|---|
| Innovus™ | Virtuoso® | MicrowaveOffice® |

Package / Board
Allegro® Package Designer / Allegro PCB

IC, PKG and Board Analysis and Signoff

DRC/LVS Timing Thermal Power SI/PI System LVS
Clarity™ Sigrity™ Celsius™
Tempus™ Quantus™ Pegasus™

*Multi-Technology Hierarchical Database*

Reduced iterations between packaging and SoC

Improves 3D-IC PPA, TAT, and productivity

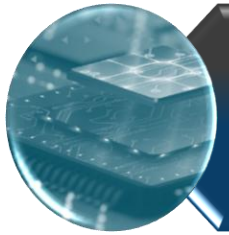

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# Integrity 3D-IC Platform – Key Areas for Innovation



## 2.5D and 3D Implementation Technology

- Innovus, Virtuoso, Allegro
- Constant PPAC improvement . 3D structure awareness



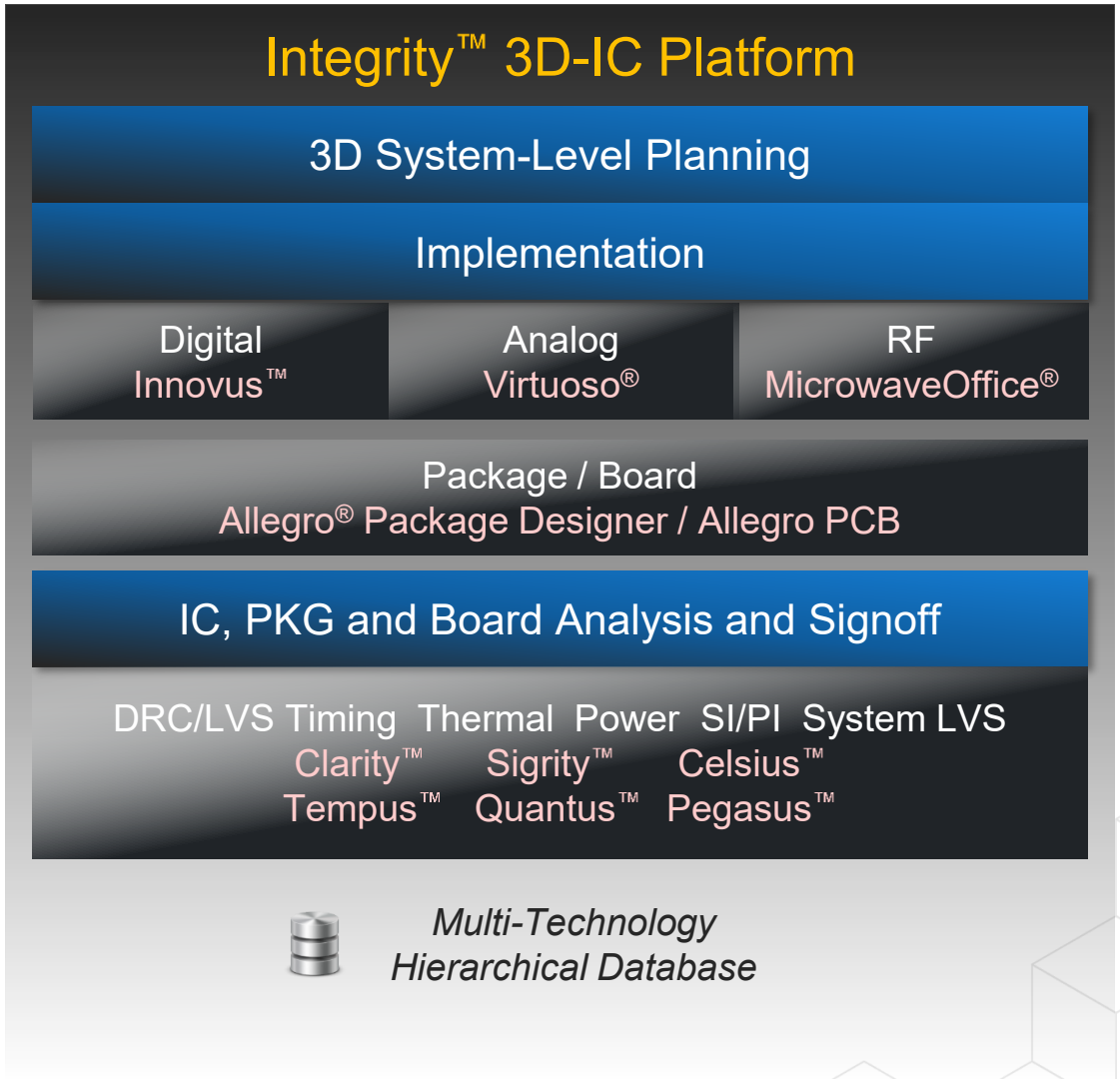
## Package and System Analysis Integration

- On-chip Timing/IR/DRC/LVS Signoff
- Off-Chip Thermal/Warpage/SI/PI Analysis



## Standards Development and Research

- Flow development with research partners
- Standards compliance and development





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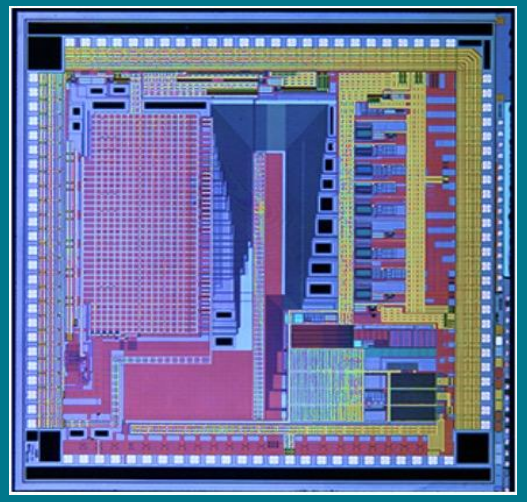


# 2.5D and 3D Implementation Technology

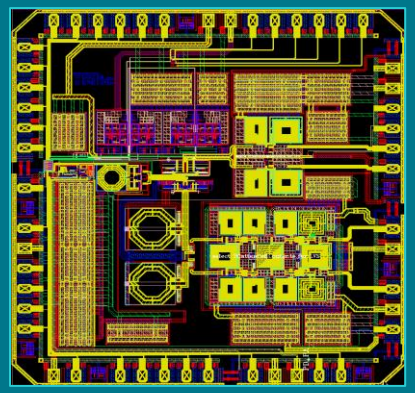

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# Integrity 3D-IC: Implementation Technologies (Innovus, Virtuoso, Allegro)

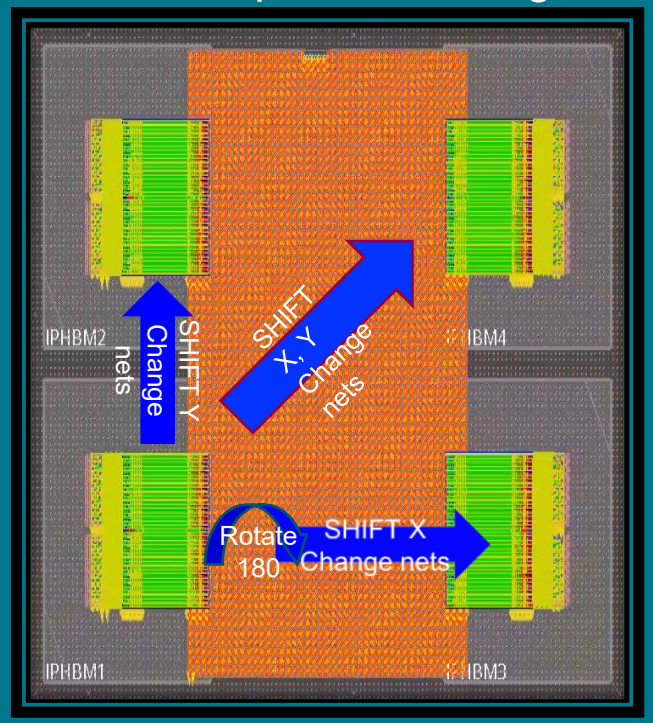
## Digital Die Routing



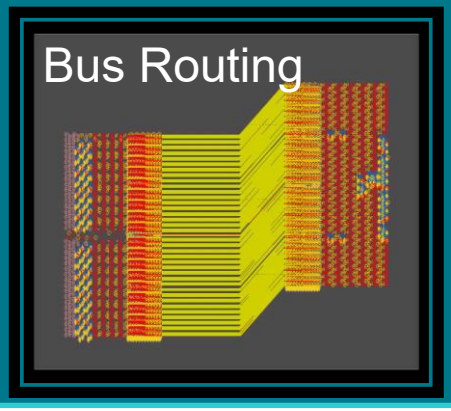
## Analog Die Routing



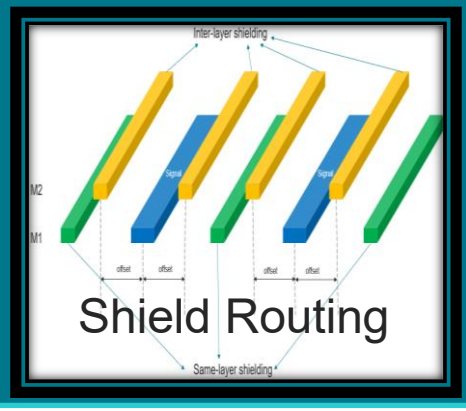
## Si-Interposer Routing



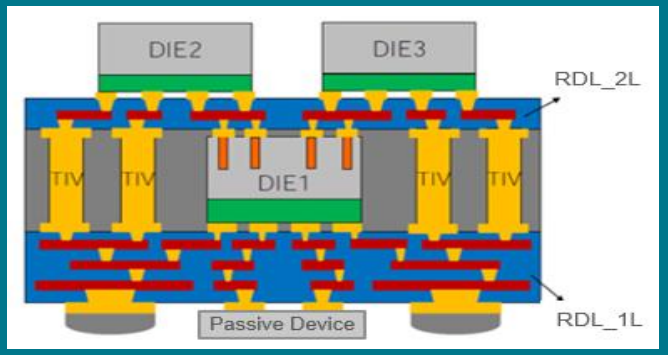

## Bus Routing

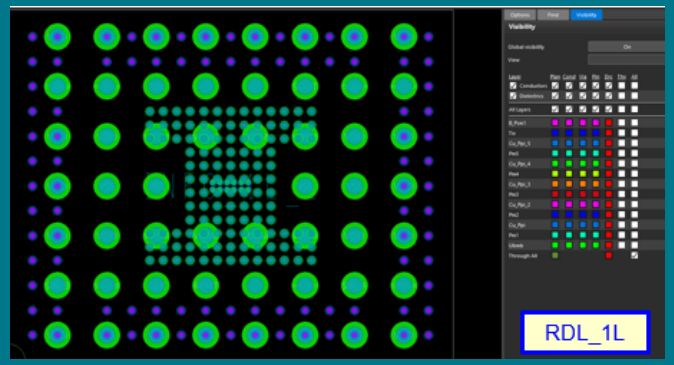


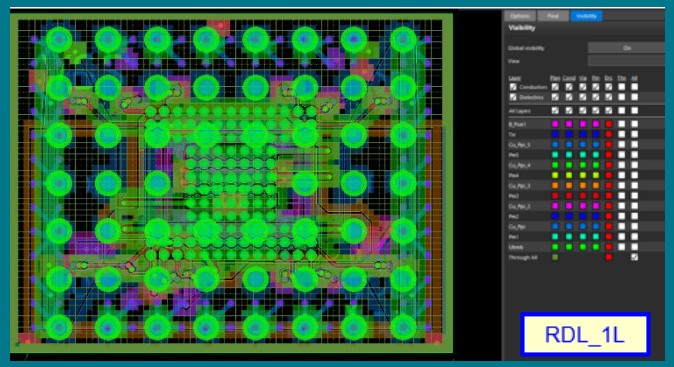

## Shield Routing




## RDL Interposer/Package Routing



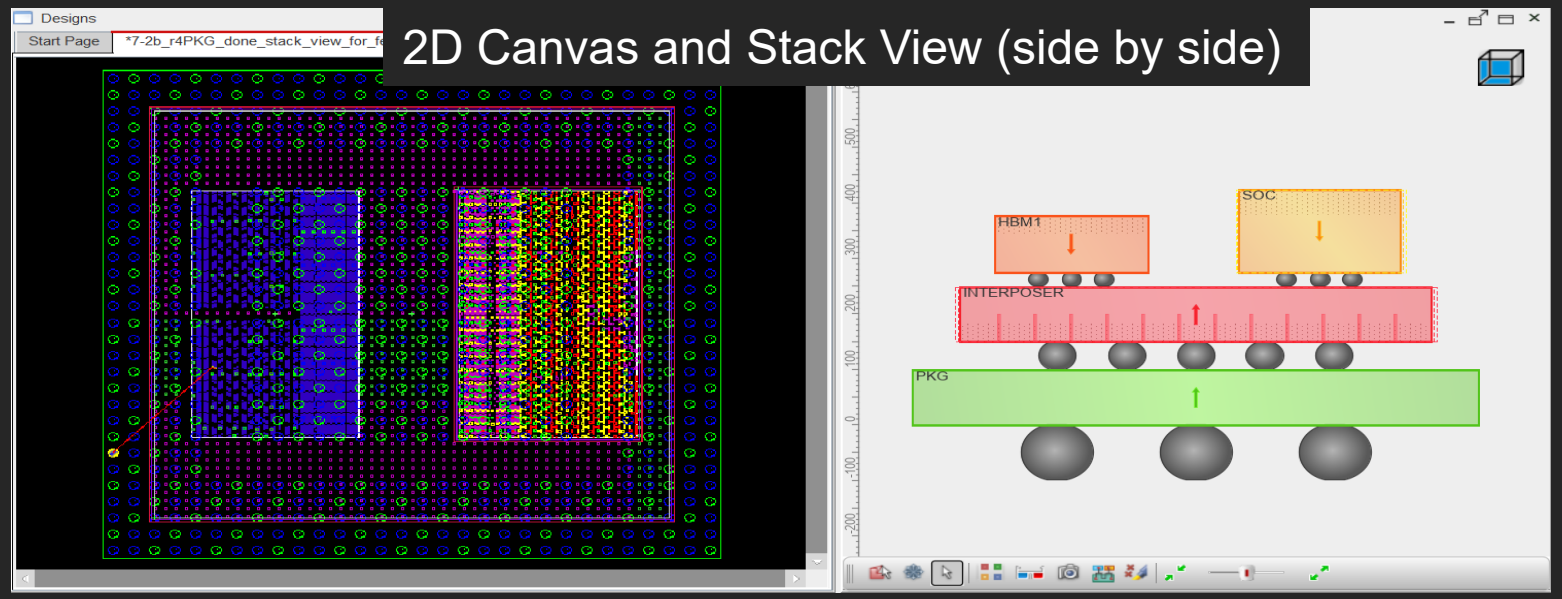





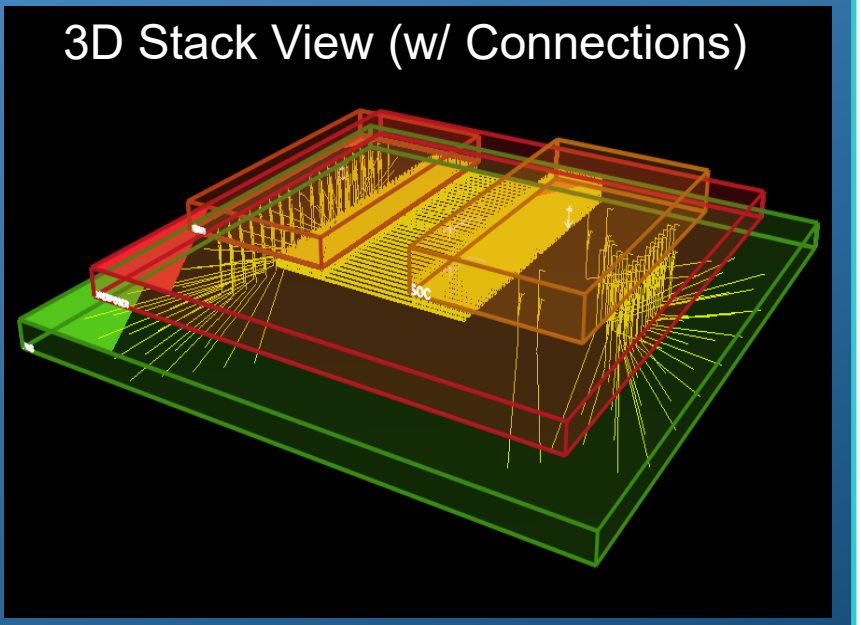
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# Integrity 3D-IC: 2D/3D Enhanced Viewing Capabilities

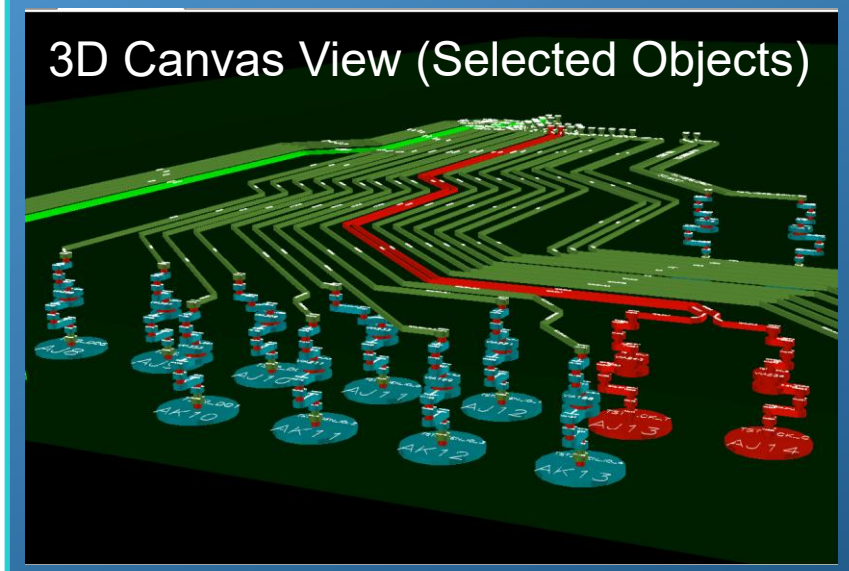
## 2D Canvas and Stack View (side by side)



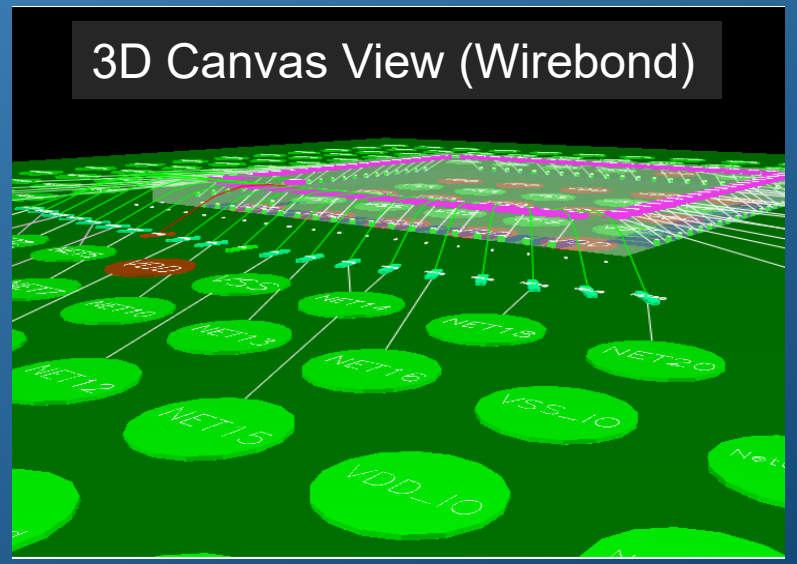

## 3D Stack View (w/ Connections)



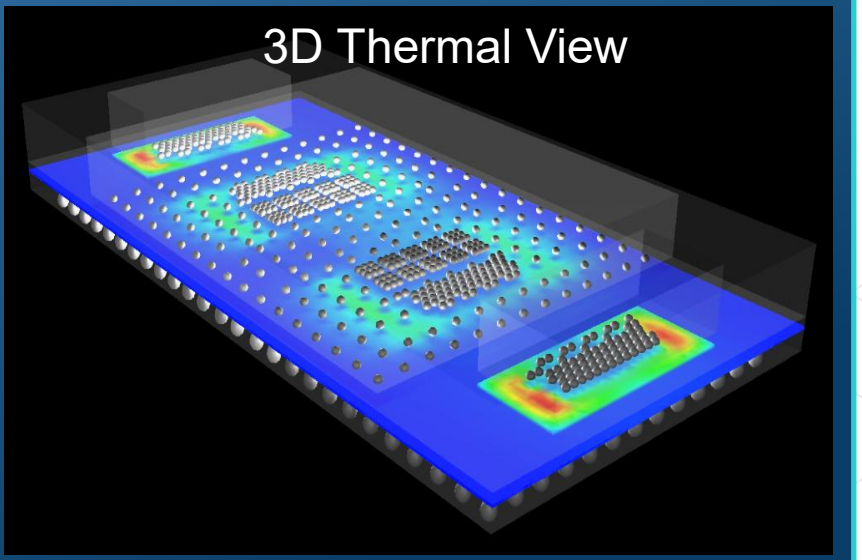
## 3D Canvas View (Selected Objects)



## 3D Canvas View (Wirebond)

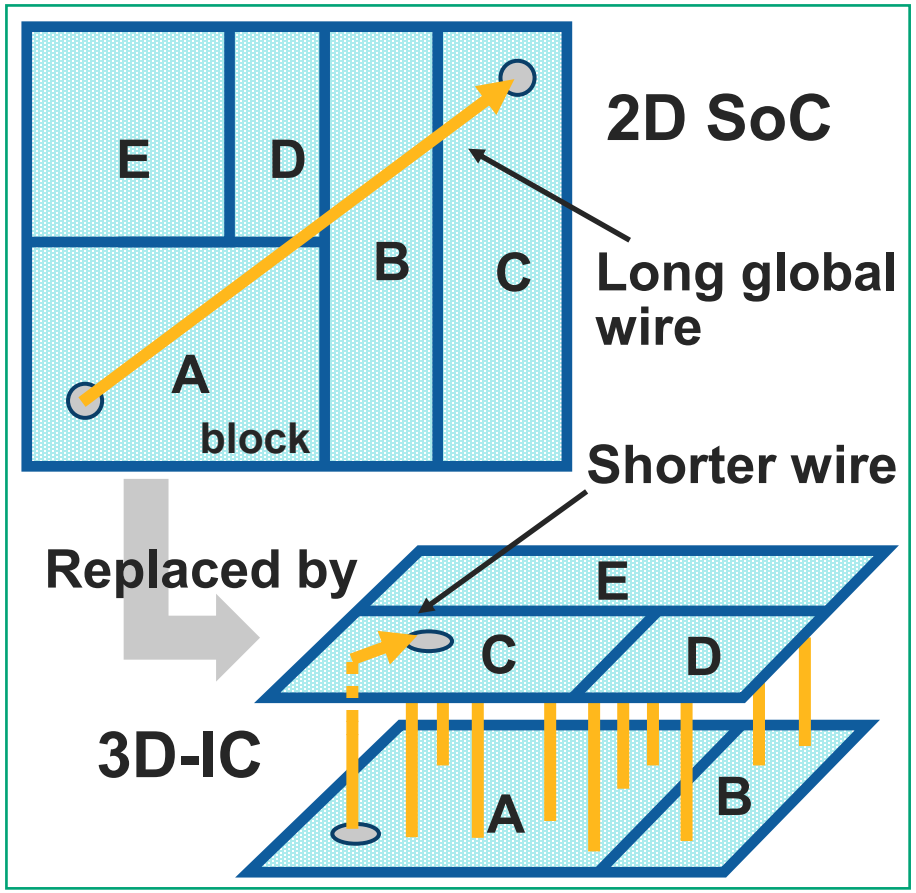


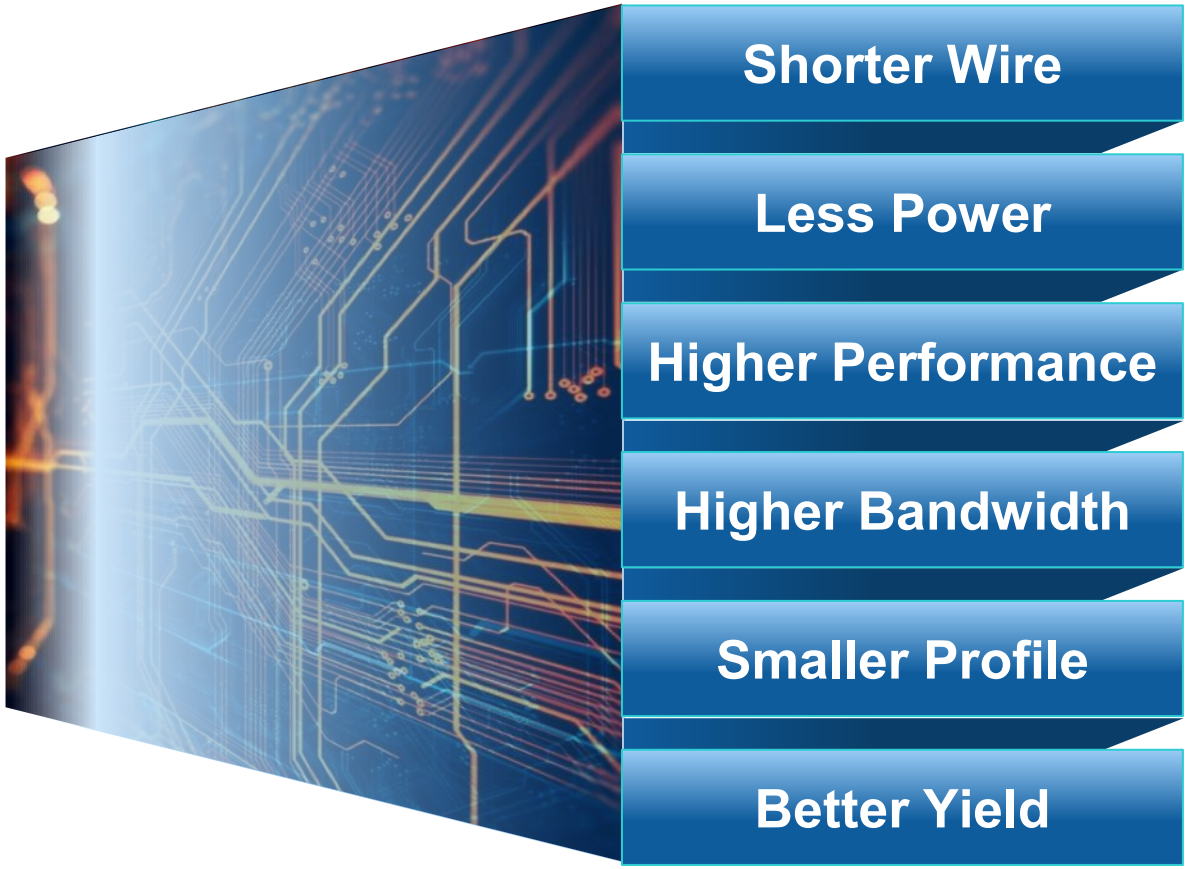

## 3D Thermal View




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# 2D/3D P&R with Early Floorplan Synthesis of Stack-Up Chips

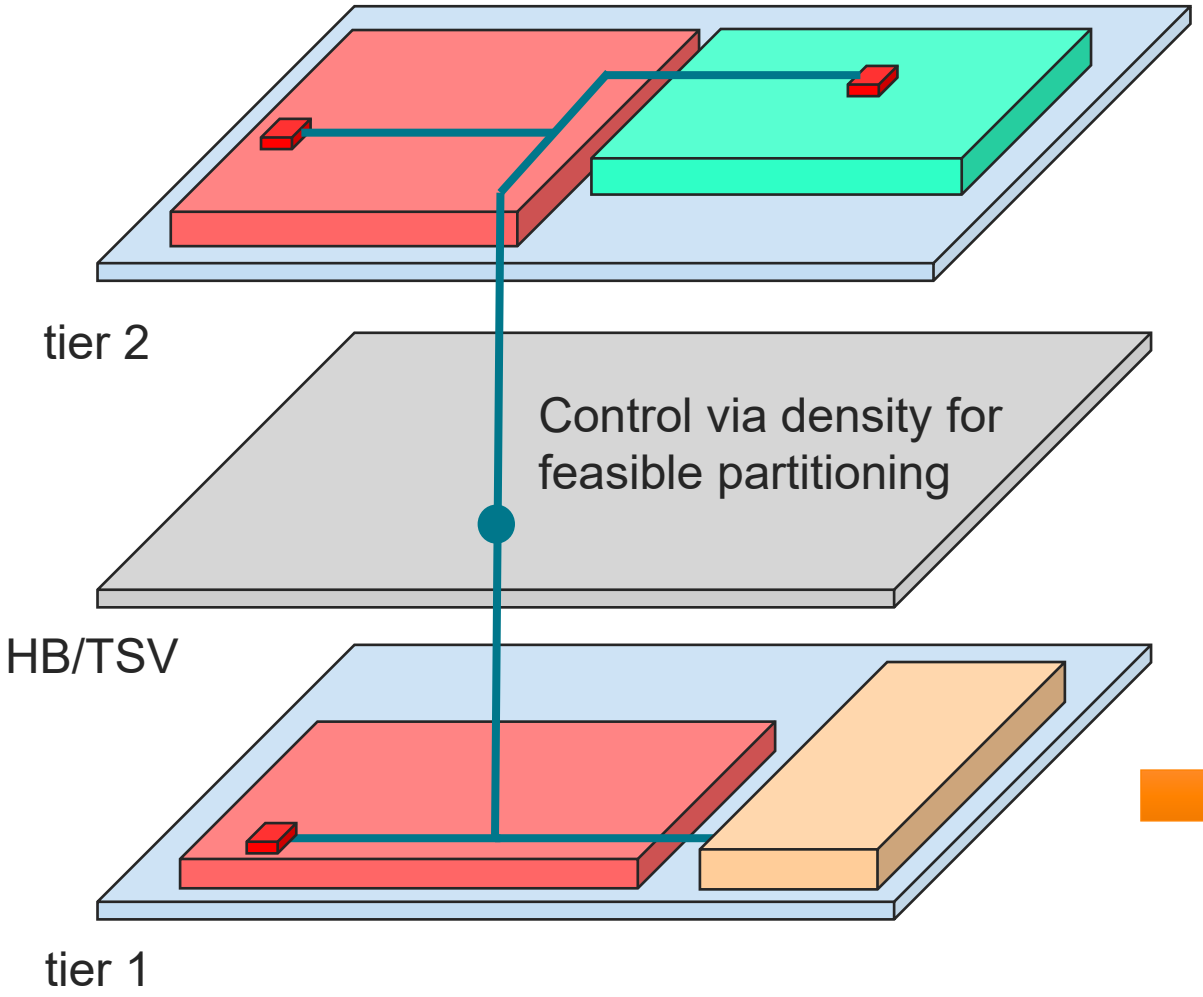


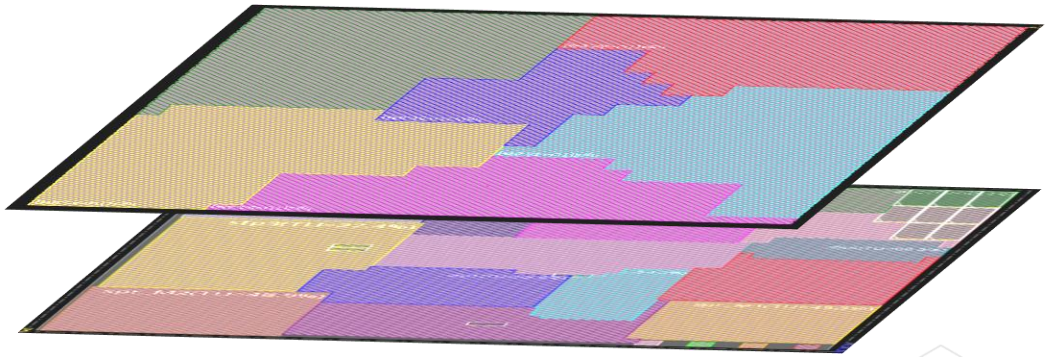

The 3D Early Floorplan Synthesis (3D-EFS) methodology is a powerful capability that enables automated rapid floorplan creation, partition shape generation, and exploration for 3D-IC design from an early stage


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# 3D Partitioning and Floorplan Synthesis



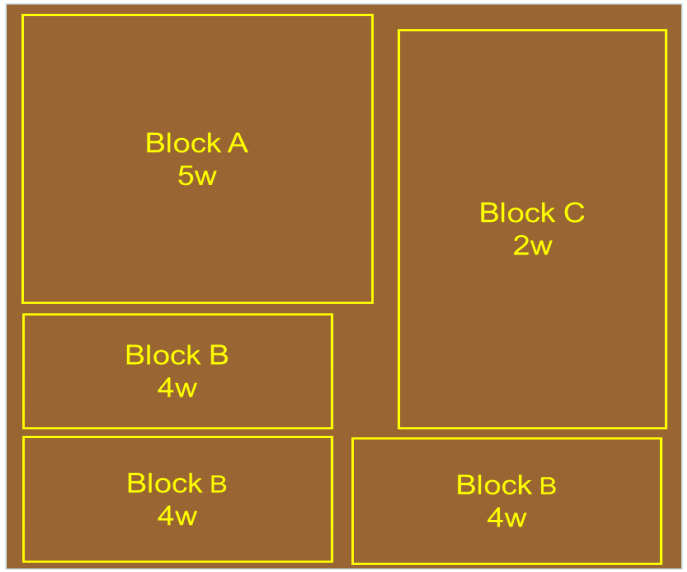

- Block-level 3D-IC integration
  - Module placer in 3D design space
  - Considering 3D wire length, number of I/O ports, and timing
  - Logic restructuring feature to divide a single netlist into multi-dies
- Generate 3D-IC die partitioning based on block-level integration
  - Better latency for critical inter-block communication




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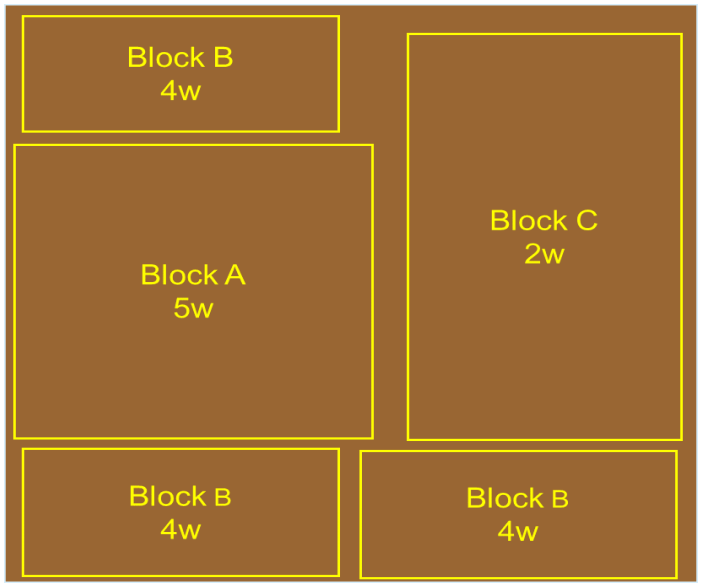
# Integrated What-If Thermal Analysis for a Floorplan

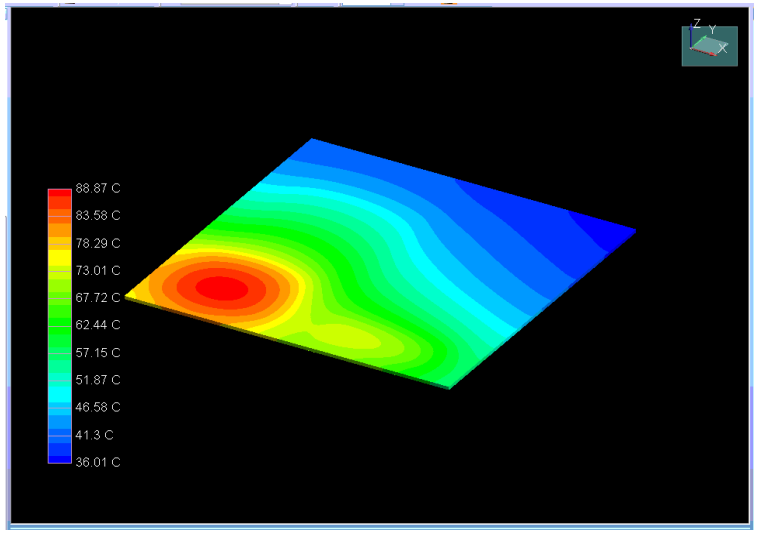
Floorplan 1




- Voltus™ power map for each IP block
- Floorplan tradeoff analysis in Celsius™ using IP thermal models
- Spreading high-density blocks reduces peak temp

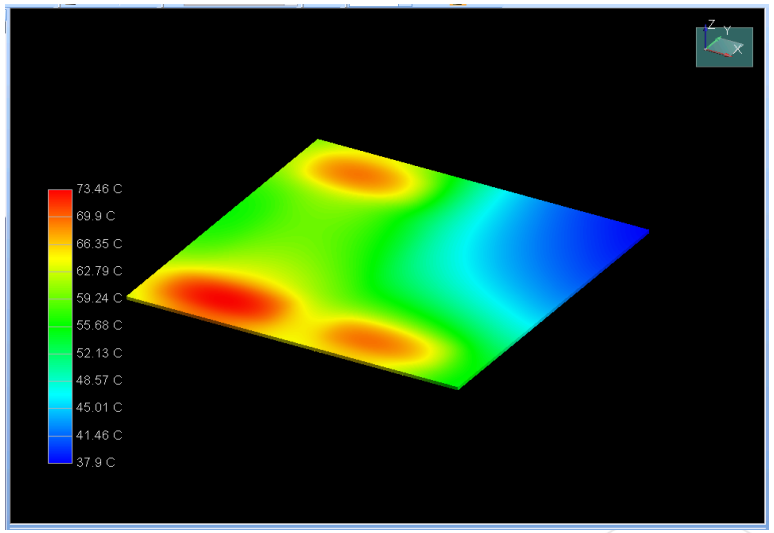
Floorplan 2




15°C reduction in max temperature

Spreading the hotspot





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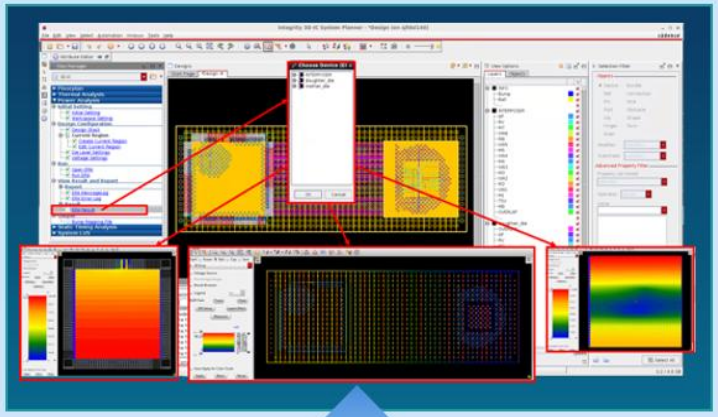
# System-Level Analysis Integration

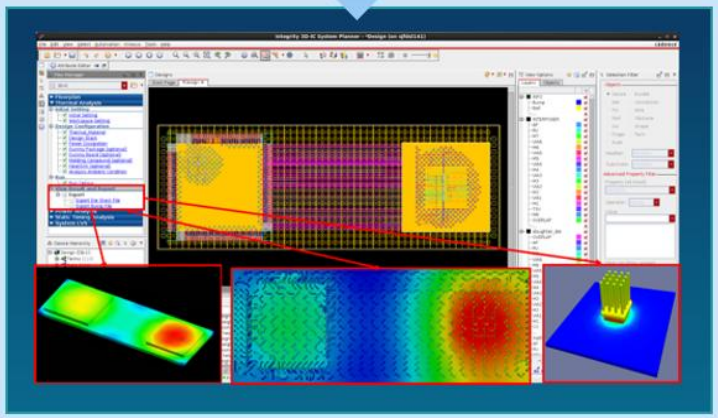
## Some new features


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# Integrity 3D-IC System-Level Analysis/Signoff Flow Integration Status

## Voltus™/Celsius™
***Early Rail (Power and Thermal) Analysis***





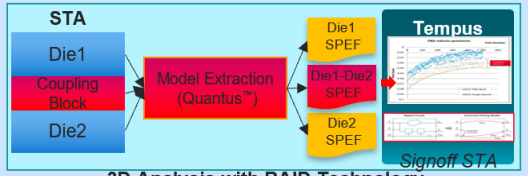
**System-Level Power and Thermal Analysis**

## Integrity™ Flow Manager
*Guided Flow*

1. Floorplan
2. Implementation
3. PDN Analysis
4. Thermal Analysis
5. Static Timing Analysis
6. System LVS
7. SI/PI Analysis

New

## Tempus™ *Timing Analysis*




3D Analysis with RAID Technology

## Pegasus™ *Physical Verification (DRC/LVS)*

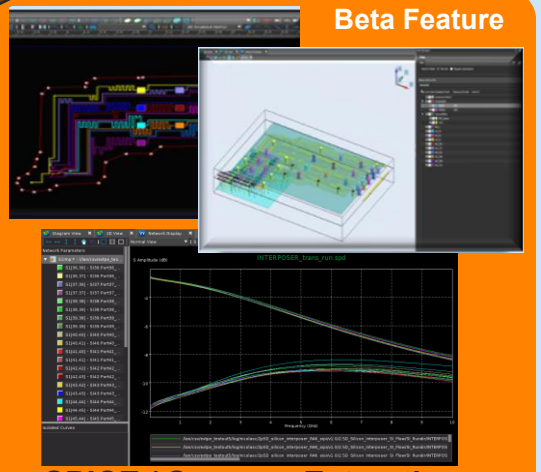


Inter-Die Connectivity Checking

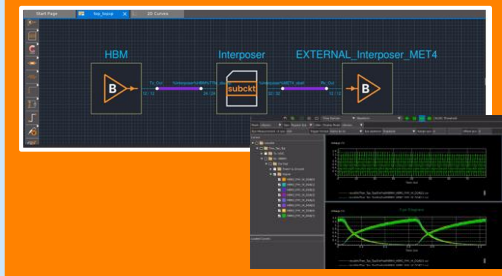
New

## Sigrity™ Tools SI/PI Analysis flows

**Beta Feature**

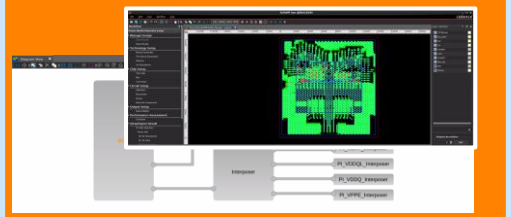


SPICE / S-param Extraction

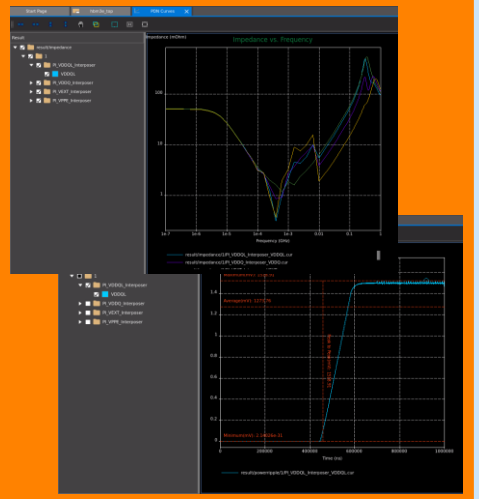



SI - Eye-diagram Analysis

**In prototyping**



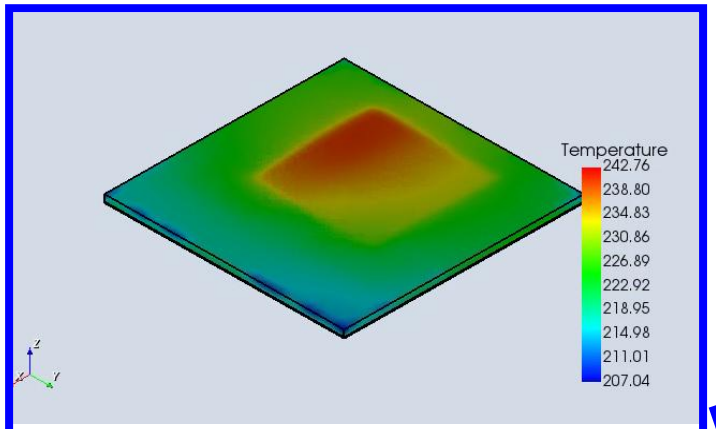
SPICE / S-param Extraction



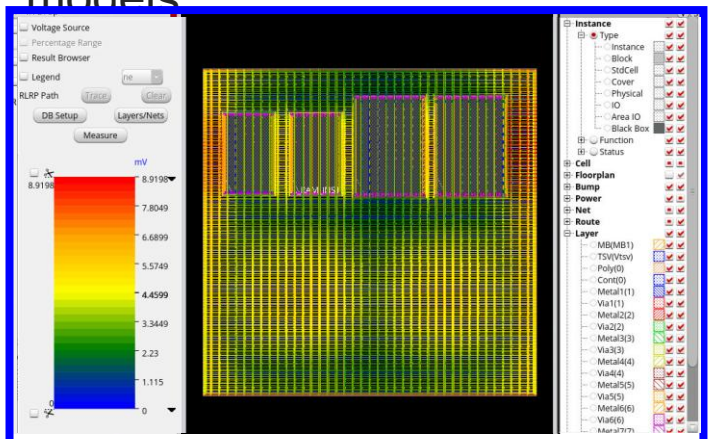
PI - Impedance and Power ripple


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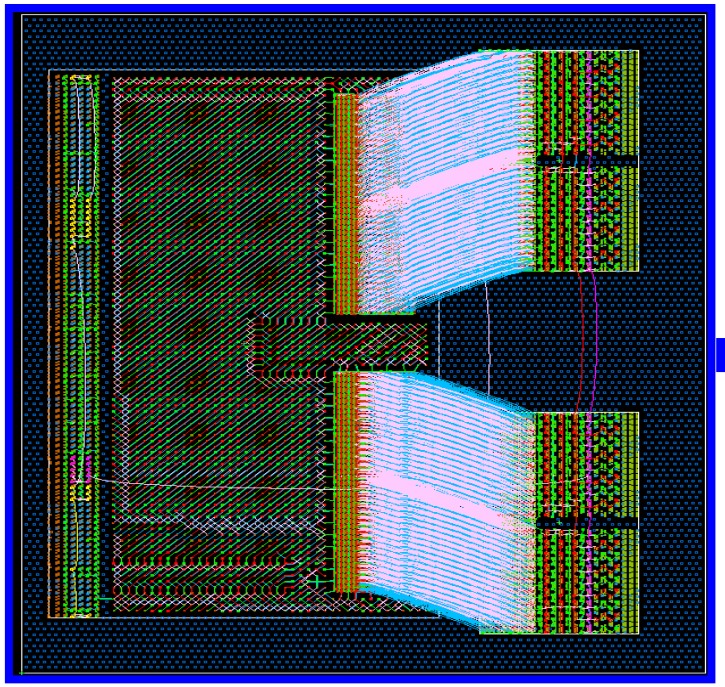
# Integration of Integrity 3D-IC with Thermal, Power, SI/PI Analysis



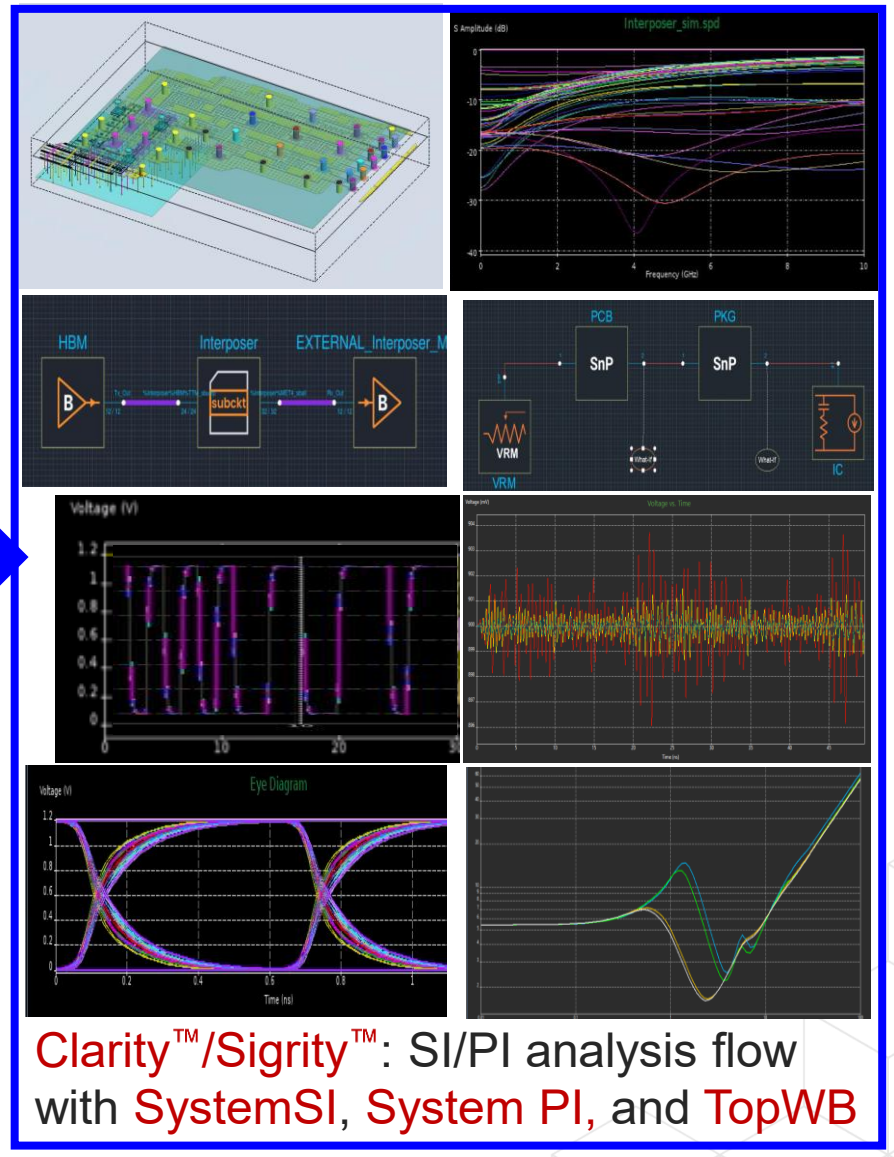

Celsius™: Early Thermal Analysis with specified power models



Voltus™: Early Rail Analysis with Chip Power Model (CPM)



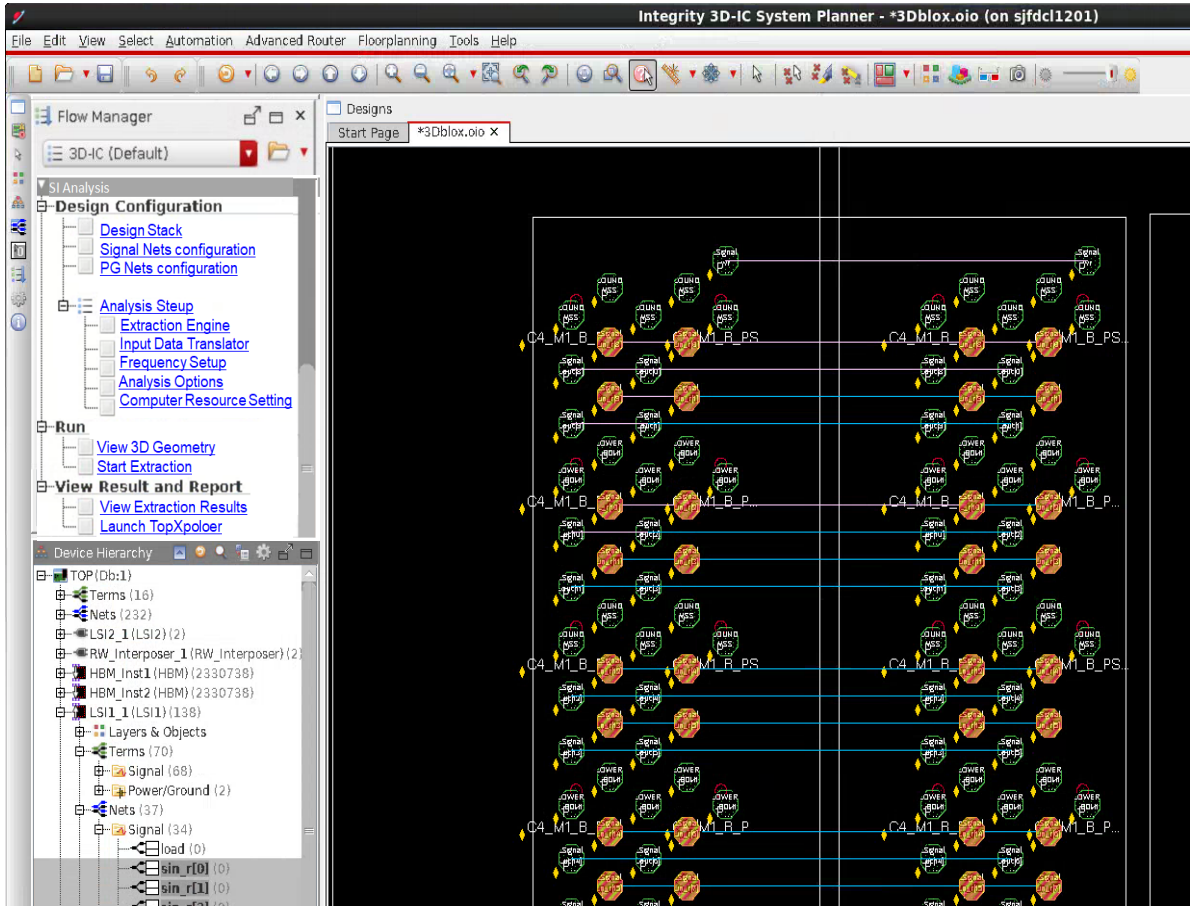
SI PI



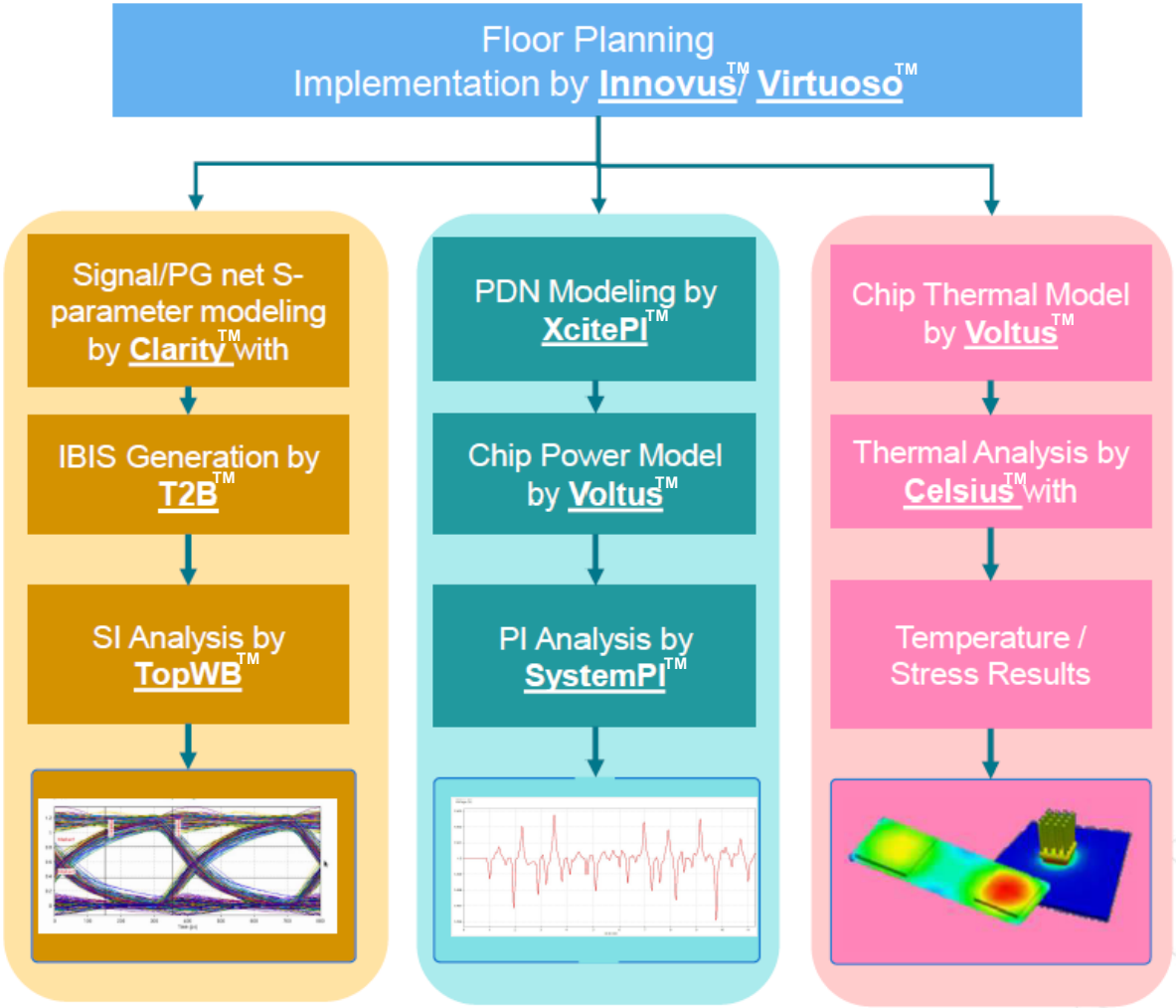

Clarity™/Sigrity™: SI/PI analysis flow with SystemSI, System PI, and TopWB


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# Integrity 3D-IC Integration Flow for SI/PI/Thermal Analyses




*An interposer sample design Integrity™ 3D-IC Platform*





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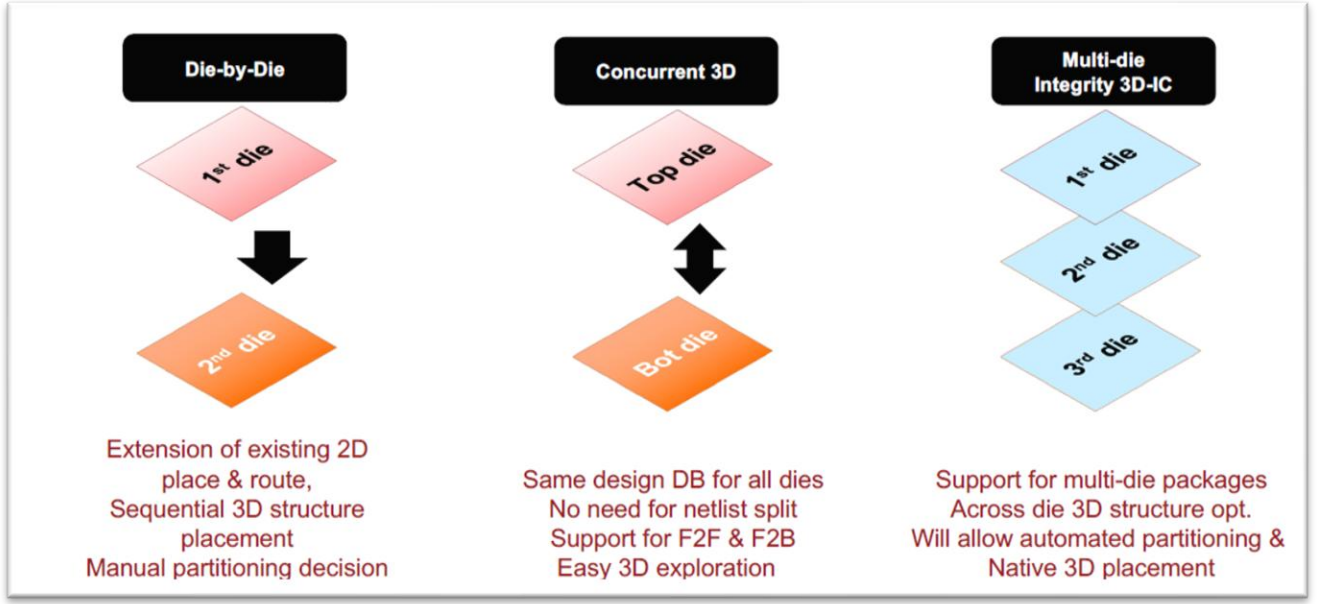
# Research Collaboration with IMEC

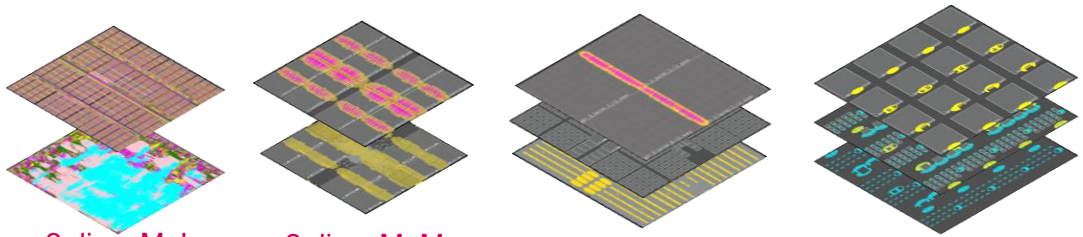
## Multi-layer stack-up flow

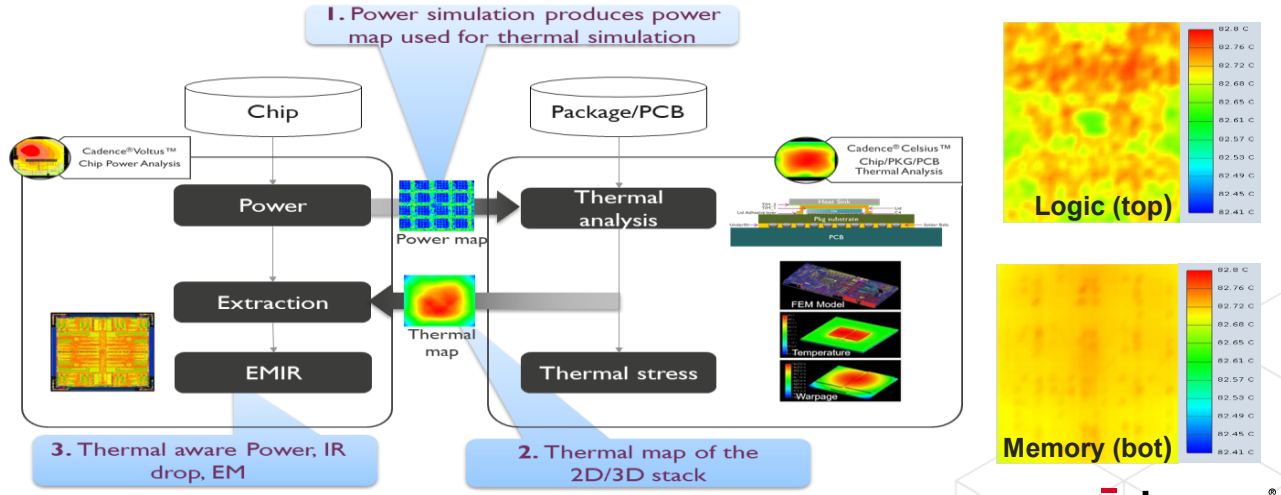

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# Research and Development for Advanced Chip Stacking Flows

- Systematic flow automation to assist research for workload simulation and configuration of 3D stacks.
- From early “Die-by-die” flow to fully automated die partitioning and concurrent 3D placement flows enabled.
- Enabled 3D Integration and PPA analysis including system level flows like full stack IR and Thermal analysis to study PPACT impact.



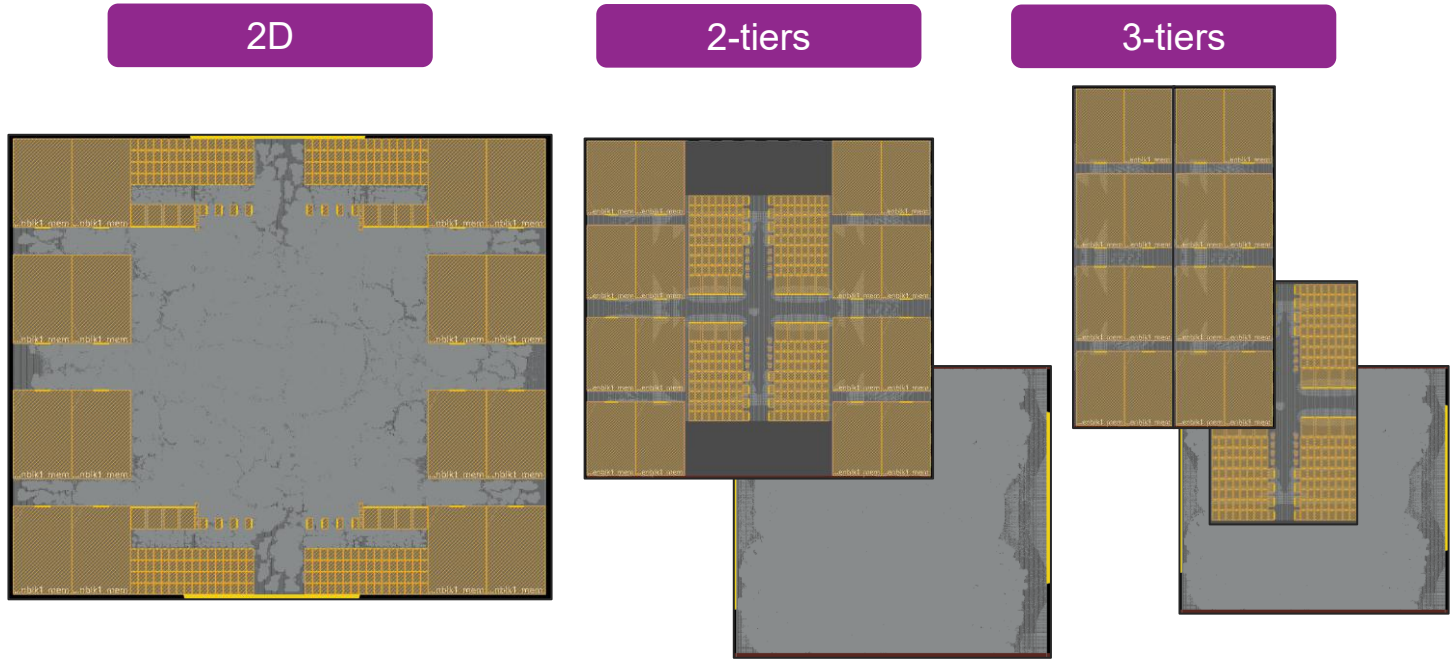





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# 2-Die/3-Die Stack-Up Based on Mempool 64-Core Layout

- Mempool design:
  - 64-core RISCV, 512KB L1, 2MB L2
  - General purpose for AI applications
  - 2D/3D implementation
  - Static power reported with 20% Switching.
  - 0.8 GHz operating frequency
- Imec A10-nanosheet PDK
  - With Buried Power Rail (BPR)
  - 3 Back-Side metals for BS-PDN
  - 0.7V & 25C
- Simulation framework (Cadence)
  - Innovus/Integrity-3DIC (PnR), Voltus (power), Celsius (thermal)
  - Total power averaged over the IC footprint
    - Detailed layout simulation in progress
  - Complete stack description




| Area | [µm²] | 1090X930 |
|---|---|---|
| Eff. F | [Ghz] | 0.8 |
| Power | [mW] | 1350 |
| Power density | [W/cm²] | 133 |

| Area | [µm²] | 2X845X820 | |
|---|---|---|---|
| Eff. F | [Ghz] | 0.8 | |
| Power | [mW] | 1190 | |
| Power density | [W/cm²] | Bot | 150 |
| | | Top | 22 |

| Area | [µm²] | 3X600X600 | |
|---|---|---|---|
| Eff. F | [Ghz] | 0.8 | |
| Power | [mW] | 1190 | |
| Power density | [W/cm²] | Top | 23 |
| | | Mid | 20 |
| | | Bot | 287 |

~ -12% power reduction impact on thermal

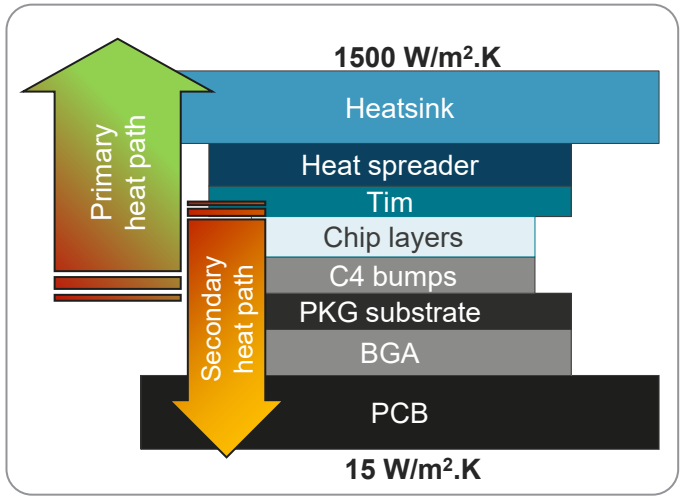
no actual PnR for 3-Dies (same data as for 2-Dies 3D-MoL)

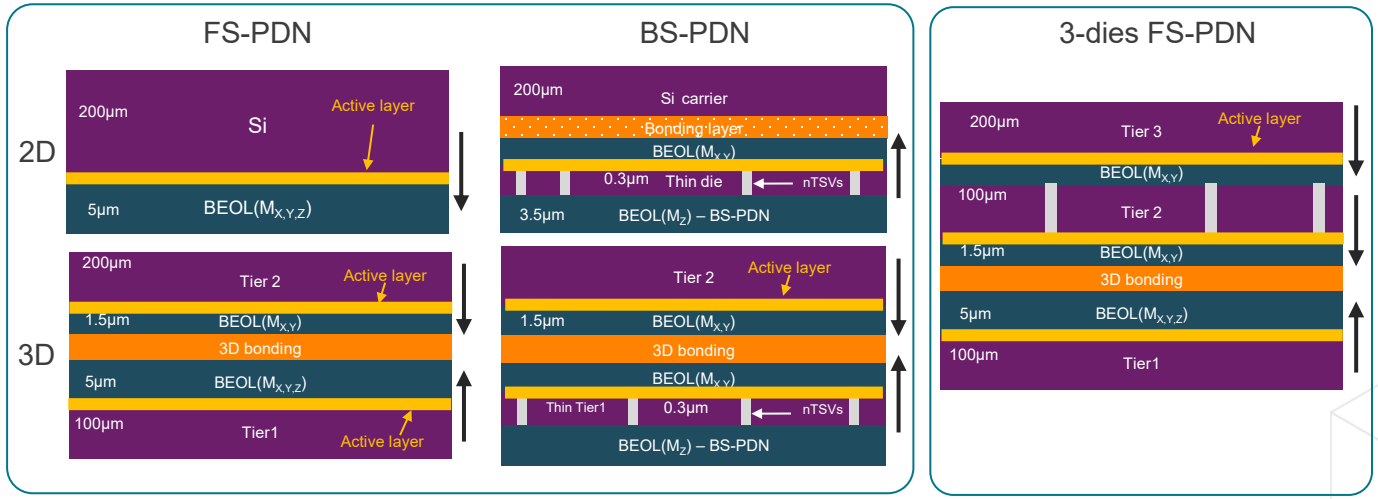
** Paper presented by Cadence in ISVLSI 2024*


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# Thermal Analysis of 3D Stacking and BEOL Technology

- Parameters /conditions under study:
  - 3D technology choice
    - Embedded micro-bumps (E-µBumps)
    - W2W Hybrid Bonding (W2W-HB)
  - Impact of 3D interface properties
    - Amount of metal in the 3D interface, metal density, depending on 3D pitch and size
  - Impact of stacking, partitioning, PDN options(Front-Side/Back-Side)
- Objective: quantifying $T_{max}$ in the stack up
- Cross-section modelling: PCB to heat-sink
- Including 3-layers stack with different die assignments (MoL, LoM, LoL, MoMoL etc.)




** Paper presented by Cadence in ISVLSI 2024*


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# Collaboration on Standards with Foundries


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# Integrity 3D-IC and TSMC 3Dblox Collaboration

**Cadence Delivers New Design Flows Based on the Integrity 3D-IC Platform in Support of TSMC 3Dblox™ Standard**

26 Apr 2023

*Highlights:*
*The Cadence® Integrity™ 3D-IC platform, a complete solution certified for all the latest TSMC 3DFabric ™ offerings, combines system planning, packaging, and system-level analysis, providing a seamless design creation and signoff flow*

*Flows fine-tuned*
*design partitionin*

*Cadence and TS*
*turnaround time f*
*and mobile applic*

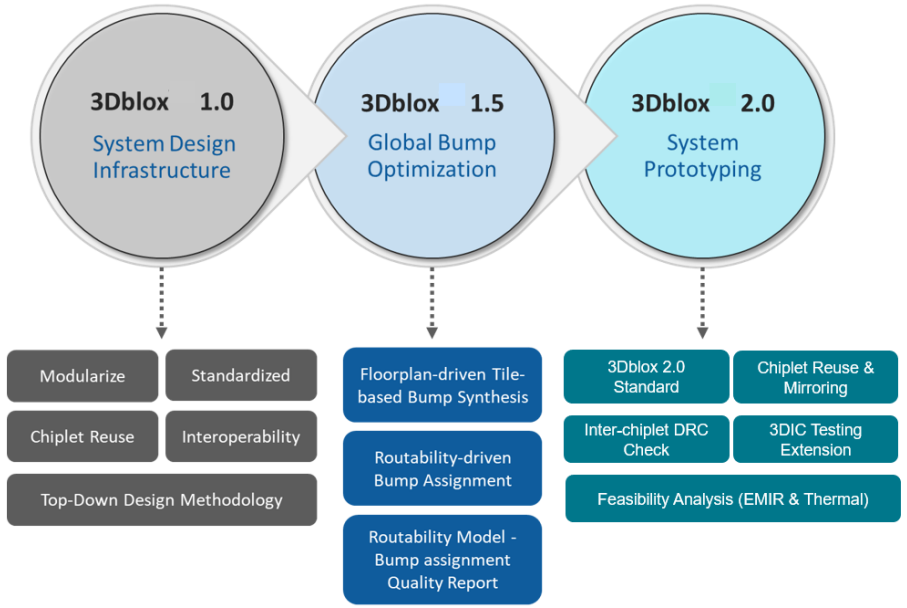
**Cadence Expands Support for 3Dblox 2.0 Standard with New System Prototyping Flows**

27 Sep 2023

*Highlights:*
*Cadence Integrity 3D-IC Platform fully supports latest 3Dblox 2.0 standard across all TSMC's 3DFabric offerings.*

*Integrity 3D-IC Platform uniquely combines system planning, implementation and system-level analysis in a single solution, enabling seamless prototyping.*

*Mutual customers creating AI, mobile, 5G, hyperscale computing and IoT 3D-IC designs can model system prototypes to accelerate multi-chiplet design turnaround time.*







http://3dblox.org


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# Integrity 3D-IC and Samsung 3DCODE Collaboration

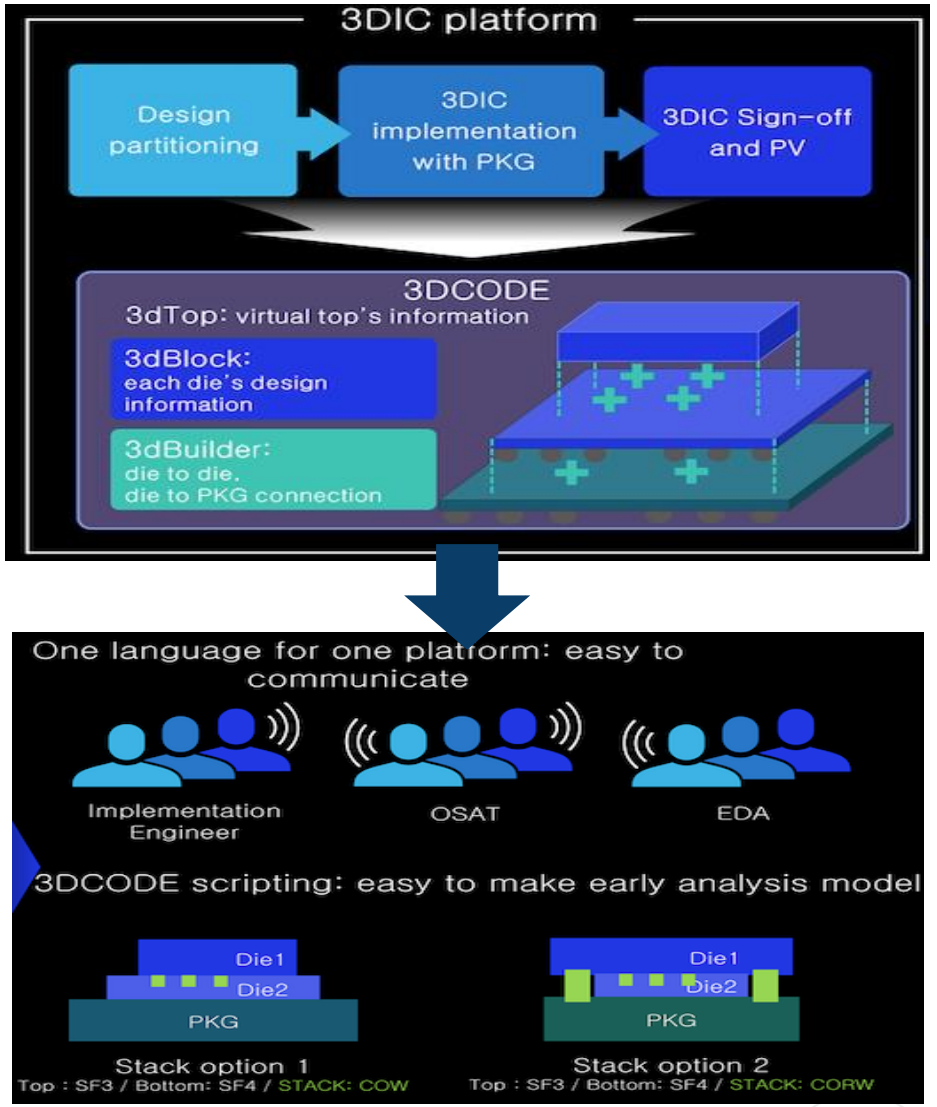
Cadence Expands Collaboration with Samsung Foundry, Providing Differentiated Reference Flows Based on the Integrity 3D-IC Platform

27 Jun 2023

*Companies advance multi-die planning and implementation, leveraging Cadence's Integrity™ 3D-IC platform, the industry's only unified platform that combines system planning, packaging and system-level analysis.*

*Integrity 3D-IC platform supports Samsung's* ***new 3D CODE standard****, enabling designers to create a variety of advanced packaging technologies.*

*Cadence and Samsung technologies provide customers with comprehensive, customized solutions for diverse configurations that reduce the overall 3D-IC design development time for 5G, AI, hyperscale, IoT and mobile applications.*





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# Integrity 3D-IC and Intel Foundry Services Collaboration

## Cadence and Intel Foundry Collaborate on Systems Foundry Enablement for the AI Era

*24 Jun 2024*

*SAN JOSE, Calif.— Cadence Design Systems, Inc. (Nasdaq: CDNS) today announced the achievement of several key milestones in its ongoing strategic partnership with Intel Foundry. Furthering the companies' collaboration on **3D-IC enablement**, EDA flows and IP development across multiple Intel process nodes, beginning with Intel 18A Cadence announced the availability of a complete **Embedded Multi-die Interconnect Bridge (EMIB) 2.5D advanced packaging flow**, enhancements to the Intel 18A digital and custom/analog flows, a wide-ranging IP portfolio and the corresponding process design kits (PDKs) across various process nodes.*

*Key milestones of the ongoing Cadence-Intel Foundry collaboration include:*

***EMIB Reference Flow:** The complete AI-driven Cadence® flow, with Integrity™ 3D-IC Platform integrating Allegro® X Advanced Package Designer ,Sigrity™ technologies, Clarity™ 3D Solver, Pegasus™ Verification System, and Virtuoso® Studio, constitutes Intel's advanced packaging reference flow that leverages its EMIB technology and is optimized to work seamlessly with Intel 18A technology. The advanced EMIB 2.5D reference flow enables customers to successfully complete full-flow heterogeneous designs, seamlessly transitioning from system-level planning, physical optimization and analysis to DRC-aware implementation and physical signoff, with unmatched productivity and time to market.*




CadenceLIVE presentations

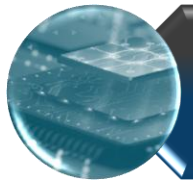

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# Integrity 3D-IC Platform – Systematic Innovation for 3D-IC Design



**2.5D and 3D Implementation Technology**

- Innovus, Virtuoso, Allegro
- Constant PPAC improvement, 3D structure awareness



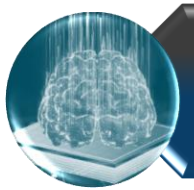
**Package and System Analysis Integration**

- On-chip Timing/IR/DRC/LVS Signoff
- Off-Chip Thermal/Warpage/SI/PI Analysis



**Standards Development and Research**

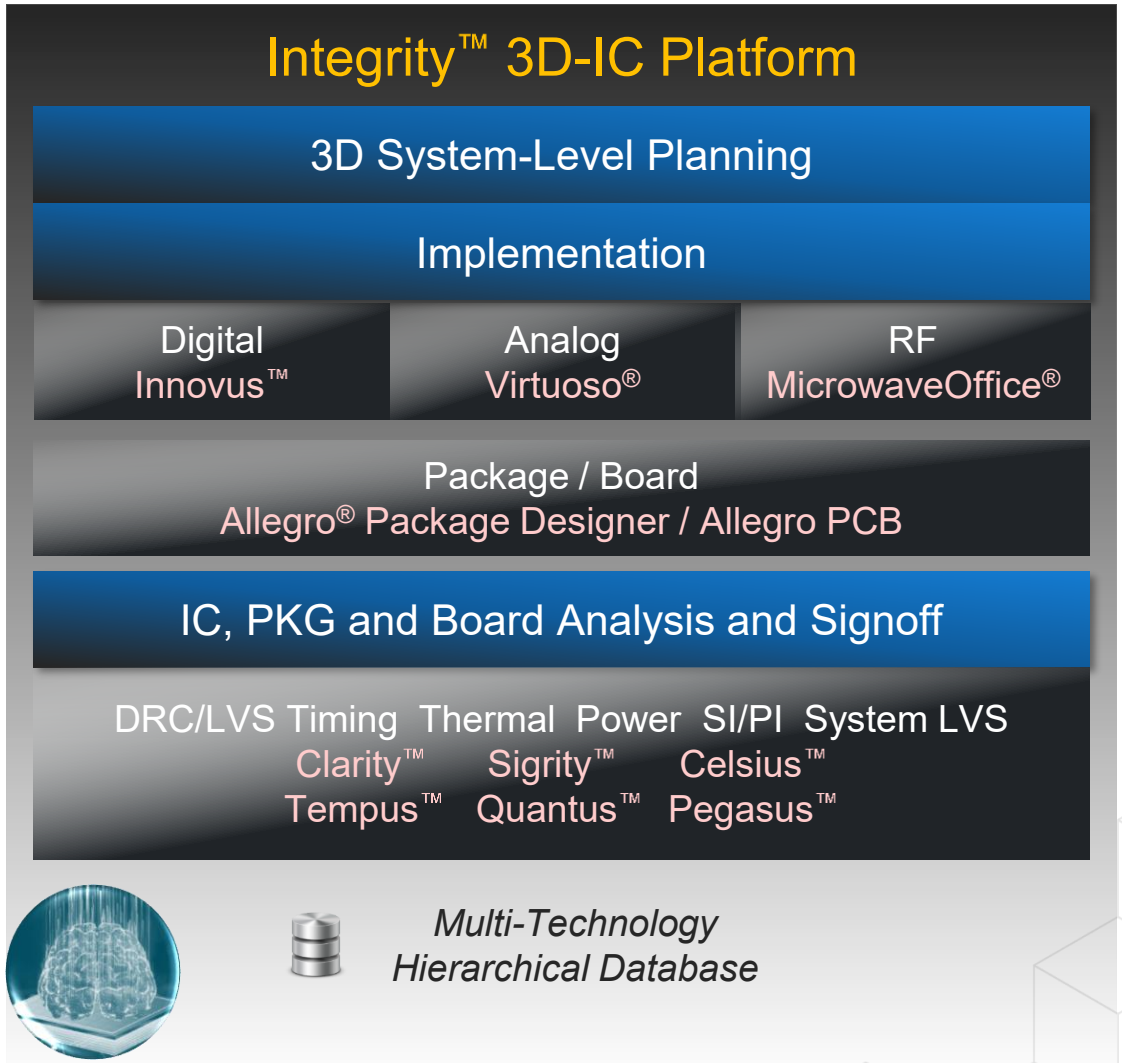
- Flow development with research partners
- Standards compliance and development



**AI to Enable and Automate Early Prototyping**

- Enable exploration, prototyping, and feasibility
- Die-disaggregation flows

**Thank You!**





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